FUNGAL ECOLOGY 4 (2011) 163–173

available at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/funeco

# Ecology of *Hericium cirrhatum*, *H. coralloides* and *H. erinaceus* in the UK

*Lynne BODDY*[a,*], *Martha E. CROCKATT*[a,b], *A. Martyn AINSWORTH*[c]

[a]*School of Biosciences, Cardiff University, Cardiff CF10 3TL, UK*
[b]*Earthwatch Institute, Oxford OX2 7DE, UK*
[c]*Mycology Section, Jodrell Laboratory, Royal Botanic Gardens, Kew, Surrey TW9 3AB, UK*

ARTICLE INFO

*Article history:*
Received 8 August 2010
Revision received 25 October 2010
Accepted 27 October 2010
Available online 17 December 2010
*Corresponding editor:* Anne Pringle

*Keywords:*
Bifactorial mating
Conservation
Endophyte
IUCN
Mycelial interactions
PCR specific primers
Red list
Tooth fungi
Wood-decay basidiomycete

ABSTRACT

*Hericium cirrhatum*, *Hericium coralloides* and *Hericium erinaceus* are rarely recorded in Europe, and *H. coralloides* and *H. erinaceus* are Red Listed in several European countries. By contrast, *H. erinaceus* is common in nature in Japan and North America and easily cultivated for edible mushroom production. The three species are saprotrophs that grow on wood of broadleaved trees, particularly beech (*Fagus sylvatica*) in Britain. How they establish in the standing tree is uncertain, but they have been detected as latently present in functional sapwood. They frequently fruit in axenic culture, perhaps indicating that lack of fruit bodies in the field truly indicates lack of mycelial individuals, although it may be that natural fruiting is impeded by encapsulation within central woody tissues. Despite prolific sporulation in wild fruit bodies, basidiospore germination on agar is poor and large progeny sets of homokaryons were relatively difficult to obtain. The duration of the homokaryotic phase is unknown but may be prolonged due to low frequency of contact between mating compatible partners in natural substrata. Homokaryotic mycelia grow at least as rapidly as heterokaryons. Both homokaryons and heterokaryons are average to good combatants, the former slightly better than the latter. Poor growth and combative ability are unlikely to be the causes of apparent rarity in Europe, but poor germination and infrequent establishment may be.

© 2010 Elsevier Ltd and The British Mycological Society. All rights reserved.

## Introduction

Many countries now have Red Lists based on IUCN (2001, 2003) criteria highlighting mostly macrofungal species of greatest conservation concern. Conservation efforts sometimes include laws to prevent picking of fruit bodies or to protect habitat in which the relevant fungus occurs, but little more (Dahlberg *et al.* 2010). To determine conservation strategies for specific fungi requires an understanding of their ecology, particularly the factors (biotic and/or abiotic) involved in their reproductive success and their decline/rarity/threats. However, there are few autecological studies on rare fungi. This paper reviews information on the distribution, occurrence, growth, reproduction and interspecific mycelial interactions of three species of tooth fungi: *Hericium cirrhatum* (=*Creolophus cirrhatus*), *Hericium coralloides* and *Hericium erinaceus*, as a case study. Stalpers (1996) recognised three other species in the white-rotting genus *Hericium* and, while acknowledging that distribution and ecological data are often lacking for the *Hericiales*, noted that some species are found fruiting on angiosperms, e.g. *H. erinaceus*, and others on gymnosperms, e.g. *Hericium flagellum*. This review highlights

* *Corresponding author.*
E-mail address: boddyl@cardiff.ac.uk (L. Boddy).
1754-5048/$ – see front matter © 2010 Elsevier Ltd and The British Mycological Society. All rights reserved.
doi:10.1016/j.funeco.2010.10.001

**Table 1 – Ecological characteristics of *Hericium cirrhatum*, *H. coralloides* and *H. erinaceus* in Britain.**

| | *H. cirrhatum* | *H. coralloides* | *H. erinaceus* |
|---|---|---|---|
| Nutritional mode | Wood saprotroph | Wood saprotroph | Wood saprotroph |
| Type of wood decay | White rot | White rot | White rot |
| Common hosts | *Fagus sylvatica*, *Quercus* spp., *Acer* spp., *Betula* spp. | *Fagus sylvatica*, *Fraxinus excelsior*, | *Fagus sylvatica*, *Quercus* spp., Acer spp., *Betula* spp. |
| Most common location of fruit bodies on wood | Fallen or cut trunks and branches; damaged parts of standing trees | Standing dead trunks, fallen trunks and larger branches | High up on wounds and stubs on living trees; fallen large diam. wood |
| Usual fruiting season | Mid-Jul.–Late Nov. | Aug.–Dec. | Aug.–Dec. |
| Duration of fruiting condition in years | 1 – few | Up to ca. 5 | Up to 20+ (possibly 40) |
| Germination in agar culture | Low | Low | Low |
| Optimum temperature for mycelial extension on agar | 25 °C | 25 °C | 25 °C |
| Optimum pH | 5.5 | 5.5 | 5.5 |
| Lowest water potential for growth | <−5 MPa | <−5 MPa | >−5 MPa |
| Mating system | ? | Bifactorial probably | Bifactorial probably |
| Combative ability | Better than heart-rotters, similar to primary colonisers – see Table 3 | Better than heart-rotters, similar to primary colonisers – see Table 3 | Slightly better than the other two species |

the known autecological information for the three target species (summarized in Table 1) and the remaining knowledge gaps (Table 2), and considers whether filling the gaps in our current knowledge will further aid conservation of these fungi in Europe.

**Table 2 – Prioritised further research requirements.**

| | |
|---|---|
| Determining numbers of genets per tree | This is the highest priority, as this will improve the species status assessments (Red List category). |
| Introgression between cultivated and wild populations of *H. erinaceus*. | To determine whether cultivated genotypes with provenances from far afield (e.g. Asia and North America) are becoming established in the wild, and the extent to which the gene pool will be altered. |
| Population biology | To confirm the mating systems of *H. coralloides* and *H. erinaceus*, and to determine the mating system of *H. cirrhatum* for which there is hardly any information. This will indicate the potential for inbreeding. |
| Mode of establishment within trees | All three species are probably latently present within functional sapwood, but more sparsely than other primary colonisers of angiosperm wood. This may be a reason for rarity. Determining how the fungi enter functional sapwood is therefore crucial to understanding rarity and might also assist beech forestry managers in their choice of trees to leave standing to ensure continuity of *Hericium* fruiting populations. |

## Distribution, conservation status and occurrence

*H. cirrhatum*, *H. coralloides* and *H. erinaceus* have all been recorded in Asia, North America and Europe (Hallenberg 1983; Ginns 1984, 1985; Boddy & Wald 2003), but there is some doubt as to whether the concepts for the three species are the same in different parts of the world. *H. erinaceus* is common in Japan and North America (Guglielmo *et al.* 2007) but each of the three species is rarely seen fruiting in Europe (Boddy & Wald 2003; Ainsworth 2004). *H. erinaceus* has been recorded from 435 localities in 23 European countries and is Red Listed in 13 of these countries (Dahlberg & Croneborg 2003). Britain and France harbour around 50 % of its known post-1980 European localities (Dahlberg & Croneborg 2003) and it is one of only four non-lichenised fungi protected by law in the UK under Schedule 8 of the 1981 Wildlife and Countryside Act. Furthermore it has a UK Biodiversity Action Plan (BAP) and is listed as of "principal importance for the conservation of biological diversity" in England and Wales in accordance with Sections 41 & 42 of the 2006 Natural Environment and Rural Communities Act. The distribution of *H. coralloides* and *H. cirrhatum* is less well documented but the former seems of greater concern and is on several Red Data Lists including those of Bulgaria (Gyosheva *et al.* 2000), Denmark (http://www.svampe.dk/ accessed 26 Jul. 2010) and Sweden (Gärdenfors 2010). It is the only one of the three species on the current, albeit unofficial, fungal Red List for Great Britain and the Isle of Man, being classed as "near threatened" (Evans *et al.* 2006 http://www.fieldmycology.net/ accessed 26 Jul. 2010). It is also rare in Germany, where it received some attention as the German Mycological Society's "Mushroom of the Year 2006".

There are two national fungal databases currently operating in Britain: the Fungal Records Database of Britain and Ireland (FRDBI) and the Association of British Fungus Groups' CATE database. The FRDBI contains less than 400 records of each species (*H. cirrhatum* 310; *H. coralloides* 188; *H. erinaceus* 365; http://www.fieldmycology.net/ accessed 14 Oct. 2010) and CATE contains less than 100 records of each species, all of

which were recorded post-1960 (*H. cirrhatum* 66; *H. coralloides* 32; *H. erinaceus* 97; http://www.abfg.org/database.php accessed 18 Oct. 2010). These records are mainly from southern and south eastern regions of the UK, especially the New Forest, Burnham Beeches, Epping Forest and Windsor Forest (Boddy & Wald 2003; Ainsworth 2004) (Fig 1). The New Forest (Hampshire) is the main national stronghold for *Hericium* and estimated to support almost 50 % of British trees with post-1960 fruiting records of *H. erinaceus*. Even here, this species is by no means frequently encountered. Targeted searches of the New Forest for fruit bodies of *H. erinaceus* in 1998, involving 32 prioritised woodlands (Wicks 1999), and in 2007, involving 49 prioritised woodlands (Ainsworth 2008), documented respectively 12 and 13 trees with fruit bodies present on the day of search, five of which were common to both surveys.

The UK sites at which the three species have been found, generally have a long history of continuous tree cover, retained dead wood, a range of tree age classes and were formerly relatively open wood pastures or hunting forests (Wicks 1999; Boddy & Wald 2003) (Fig 1 and 2). All the distribution maps (Fig 1) show increases in range of *Hericium* spp. post-1960 and, of the FRDBI records of the genus made between 1813 and 2009 (http://www.fieldmycology.net/ accessed 27 Jul. 2010), 50 % originated during the 15 yr since Oct. 1994. However, increases probably do not reflect increases in population size. Instead, they serve to highlight the tremendous increases in local recording group activity, database usage, networking and publicity, all of which present challenges to assessments of underlying population trends. Note also that the numbers of records in databases should not be taken as absolute values, as not all finds are reported and data sharing networks generate replicate records, but the data are valuable for comparison between the three species and between occurrences on different types of substratum.

Although they are sometimes seen fruiting on living trees (particularly *H. erinaceus*), more detailed investigation usually indicates that it is the dead parts of such trees that harbour the subtending mycelium as all are saprotrophs of dead wood. Descriptions of *Hericium* species, and *H. erinaceus* in particular, as parasites (e.g. Hansen & Knudsen 1997) or weakly parasitic (e.g. Legon & Henrici 2005) lack experimental support, and might simply represent the widely-perceived equivalence of "parasitic" and "fruiting on living trees". There appears to be selectivity (shown by reproducing individuals at least) for particular tree species in Britain. Of the 852 *Hericium* records held on FRDBI (accessed 25 May 2010), 576 had accompanying tree data (tree genus or species). Of these, about 80 % were on beech (*Fagus sylvatica*), about 8 % were on oak (*Quercus*) – none

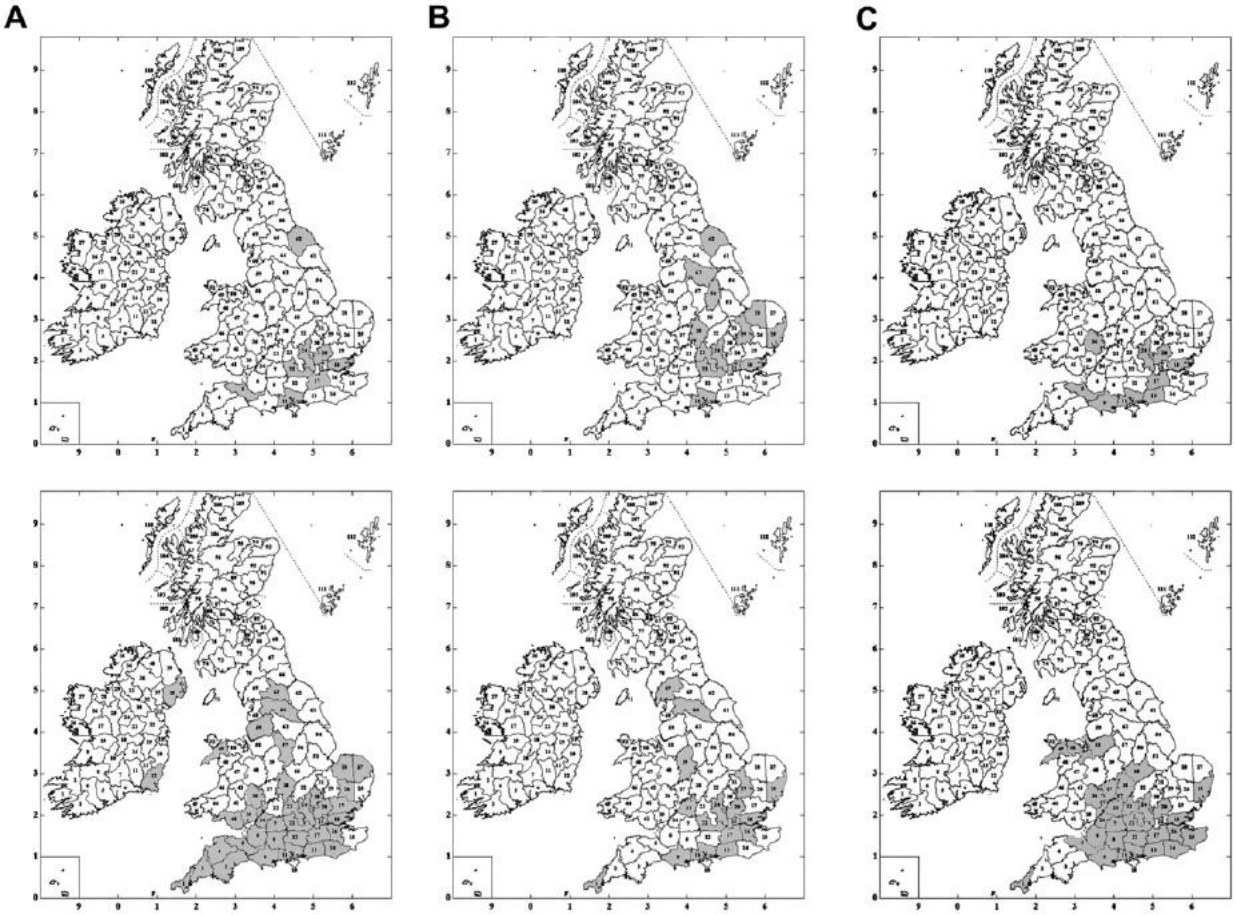


**Fig 1 – Maps showing the vice (botanical) counties in which British and Irish records of (A) *Hericium cirrhatum* (B) *Hericium coralloides* and (C) *Hericium erinaceus* have been made pre- (top row) and post-1960 (lower row) based on FRDBI (http://www.fieldmycology.net/) & CATE (http://www.abfg.org/database.php) databases accessed 26 Jul. 2010.**

**Fig 2 – Fruit bodies of *H. cirrhatum*, *H. coralloides* and *H. erinaceus:* (A) mature *H. cirrhatum* fruit body on felled beech trunk in former wood pasture (Burnham Beeches) above a 25 square-well spore trap plate (100 cm$^2$); (B) immature *H. cirrhatum* fruit bodies developing on beech slice in the laboratory; (C) *H. coralloides* fruiting on beech snag in pony-grazed woodland (New Forest); (D) *H. coralloides* fruit body with snow on fallen beech trunk in deer park (Windsor Great Park); (E) *H. erinaceus* fruit body emerging from branch stub of live beech (New Forest); (F) *H. erinaceus* fruiting (arrowed) on oak pollard branch stub (Epping Forest); (G) *H. erinaceus* fruiting in the laboratory with white spore deposit (arrowed) on agar slope below; (H) *H. erinaceus* fruiting in Jan. from centre of sawn Turkey oak (Windsor Great Park); (I) row of *H. erinaceus* fruit bodies on central wood of beech branch that fell 1–2 yr previously (New Forest). Fruiting had also occurred on corresponding trunk scar. Photo credits: A & G © Martha Crockatt; B, © Juliet Hynes; C–E, H & I © Martyn Ainsworth; F © Andy Froud. Scale bars are approximate.**

of which were of *H. coralloides*, and about 5 %, mostly of *H. coralloides*, were on ash (*Fraxinus excelsior*). The remainder comprised a few records on other trees such as maple/sycamore (*Acer*) and birch (*Betula*) (Fig 3; accessed 26 Jul. 2010).

All three species are found on both standing and fallen wood, usually on trunks and large branches (Boddy & Wald 2003) (Fig 4). *H. erinaceus* is typically found fruiting high up on wounds, or on broken branch stubs of living trunks, or on

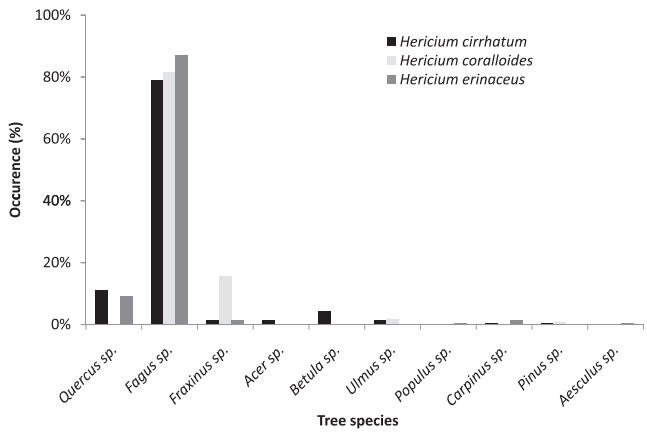


**Fig 3 – Number of records of *H. cirrhatum*, *H. coralloides* and *H. erinaceus* in the FRDBI (accessed 26 Jul. 2010) with respect to tree species.**

fractured or fallen large diameter (>10 cm) wood. *H. coralloides* fruits most commonly on standing dead trunk remains (snags), or fallen trunks and branches, whereas *H. cirrhatum* is usually found fruiting on fallen or cut trunks and branches, but also on damaged parts of standing trees, sometimes high off the ground (Marren & Dickson 2000; Boddy & Wald 2003; Ainsworth 2004) (Fig 2).

## Fruiting, spore production and dispersal

The fruiting season of *H. cirrhatum* usually begins in mid-Jul. in the UK and fruiting often continues until late Nov., while *H. coralloides* and *H. erinaceus* tend to fruit between Aug. and Dec. (Fig 5). Fruit bodies of the latter two species are occasionally found surviving and sporulating in Jan., or even Feb., having withstood periods of frost and snow. *H. erinaceus* has been observed to fruit intermittently on the same tree for 2 decades or more whereas *H. coralloides* seems to fruit more regularly on a tree but for fewer seasons, and *H. cirrhatum* fruits irregularly and rarely in the same place for two seasons running (Marren & Dickson 2000; Ainsworth 2004). *H. cirrhatum* and *H. erinaceus* can fruit on relatively undecayed

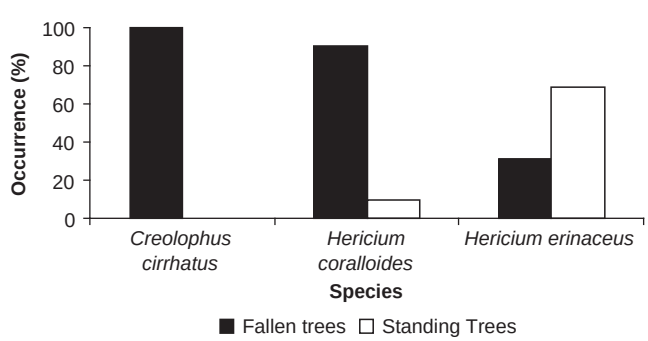


**Fig 4 – Occurrence of fruit bodies of *Hericium cirrhatum*, *Hericium coralloides* and *Hericium erinaceus* on standing (white bars) and fallen trees (black bars). From Boddy & Wald (2003).**

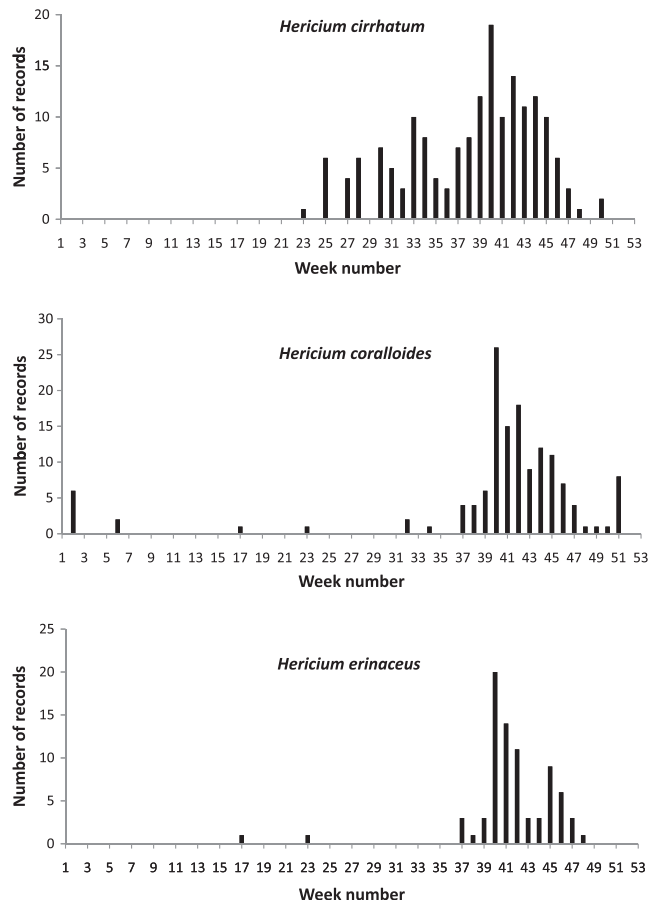


**Fig 5 – Time (week 1 = first week of Jan.) when *H. cirrhatum*, *H. coralloides* and *H. erinaceus* were reported fruiting in the FRDBI (accessed 26 Jul. 2010). Records have only been included when the actual day was specified.**

wood whereas the fruit bodies of *H. coralloides* usually signify a relatively advanced state of underlying decay.

All three *Hericium* species fruit rapidly in artificial culture, indicating that fruiting late in life is not intrinsic. Field observations of fruiting *H. erinaceus* on sawn and fractured trunks and recently fallen branches (Fig 2H and I) strongly suggest that it inhabits the central, dysfunctional core of large-diameter wood. Once "released" from the "imprisoning" envelope of sapwood functional in water conduction, such naturally-occurring centrally-located mycelia can produce large fruit bodies within 1–2 yr of being exposed to the air (Ainsworth 2008). Beech that has grown in an open habitat away from close neighbouring trees, e.g. in parkland and wood pasture, is subjected to increasing fractures and loss of large lower branches with age. Thus, the observed tendency for this fungus to fruit late in life on old trees may be a consequence of the increased opportunities for the mycelium to reach and fruit from an exposed wood surface.

Spore dispersal is essential for *Hericium* establishment in new resources and, when coupled with outcrossing, it is also important for increasing genetic diversity of populations. Understanding spore production, dispersal and germination is therefore crucial to understanding a species' population

dynamics, genetics and distribution patterns (Edman & Gustafsson 2003). The distance to which spores of the three *Hericium* species were dispersed in the field was investigated using species- and spore-specific traps (Crockatt *et al.* 2007) (Fig 2A). With this method, agar plates (either as individual plastic dishes or partitioned into square wells) are pre-colonised by single-spore-derived mycelia (primary mycelia) of the target species (previously demonstrated to be homokaryotic with detectable mating reactions regulated by a multiallelic outcrossing mechanism). These dishes are then placed in the field in the vicinity of the target fruit body and at different distances from it, in this case at 10 and 100 m in the four cardinal compass directions. The rationale is that basidiospores that land on the plates are likely to be of dissimilar mating type to the resident homokaryons and will therefore be mating compatible (Adams *et al.* 1984). In many basidiomycetes, including *Hericium* spp., successful mating can be detected by the stable production of clamp connections on hyphae in the plates or, preferably, on subcultures.

A high proportion of the wells in spore trap plates positioned directly beneath the target *Hericium* fruit bodies developed clamp connections. The proportion of wells with clamps (i.e. mating detected) decreased with distance from the fruit body though some mating of test mycelia was still detected at 100 m from the source fruit body. Unopened control plates were also left in the field near target fruit bodies, but all lacked clamp connections (no detectable mating) following laboratory incubation. The three *Hericium* species appear to be no different from the majority of wood-decay basidiomycetes wherein most spores fall within 1 m of the parent fruit body, with very few travelling further than 100 m and only occasionally reaching 1000 km (Hallenberg & Küffer 2001).

Before contributing to the mycelial and potentially breeding population, a spore must germinate, which may require specific conditions (see references in Merrill 1970). Germination of basidiospores, collected by placing small pieces of sporulating fruit body on glass slides overnight in Petri dishes to deposit spore prints (10 fruit bodies from *H. cirrhatum*, 5 from *H. coralloides* and 16 from *H. erinaceus*), was evaluated by spreading suspensions onto agar media at a density of about 40 spores/field of view at 100× magnification. Various biotic and abiotic conditions were tested including malt (1 % and 2 %) agar; agar that contained charcoal, cellulose or beech sawdust; and agar with a plate colonised by a conspecific homokaryon or wood isolate (heterokaryon) taped above the test plate at 20 °C in darkness (Crockatt *et al.* 2007). Germination was poor for all species, usually less than 1 % after 20 weeks. However, in 8 of the 75 treatments/collections plated from fresh spore collections, germination was more extensive, though there was nothing strikingly different from other treatments/collections. Time to germination of *H. coralloides* and *H. erinaceus* was inversely related to spore age, taking on average about 8 d to germinate from fresh collections and 60 d to germinate if spores had been kept for 24 weeks, though there was considerable variation, and germination was consistently less than 1 %. However, even though germination was low after 60 d, these data suggest that spores could remain dormant while environmental conditions are unfavourable. Further, even older spores may still be viable.

Low germination of basidiospores of these species may be a factor contributing to the rarity of recorded fruit bodies in the field. However, the laboratory incubation conditions used may not have been optimal for germination. Even though germination may be low in the laboratory, the spore trapping experiment provided evidence that germination can occur frequently or rapidly in the field when homokaryons are present.

## Growth and establishment

Initially, mycelia of *Hericium* spp. derived from single basidiospores are homokaryotic; this stage of the basidiomycete life cycle is generally considered to be short-lived (e.g. Kauserud *et al.* 2006) although there is little experimental data on how long it persists in the field. Homokaryons of some wood-decay species e.g. *Trametes versicolor* and *Heterobasidion annosum*, can persist for several years in the field (Coates & Rayner 1985; Garbelotto *et al.* 1997; Redfern *et al.* 2001). Also, for a rare species, mating may be delayed if there are only low numbers of potential mates in the environment, and this is reflected in the field by an inverse relationship between number of mycelia of a species and the number that are homokaryons (Stenlid 1994). The homokaryotic phase may therefore be of greater importance for rare species than for common species, as it is likely to last longer. Most studies on the effects of abiotic variables on mycelial extension are conducted on heterokaryotic mycelia but, clearly, this is not necessarily sufficient.

The extension rates of *H. coralloides* homokaryons and heterokaryons have been compared in agar culture (15 homokaryons from 3 fruit bodies, 5 heterokaryons from fruit body tissue isolation, and 10 artificial heterokaryons subcultured from pairs of mated sibling and non-sibling homokaryons at 20 °C) (Crockatt *et al.* 2008). There were some significant differences between extension rates of the homokaryons and heterokaryons but no consistent trends. For all three *Hericium* species' growth on agar, the cardinal points for temperature are similar to those of other wood-decay basidiomycetes common in northern Europe (Rayner & Boddy 1988), with an optimum temperature for radial extension at 25 °C (3.3–6.7 mm $d^{-1}$) with slow growth at 5 °C and 10 °C and no growth at 35 °C (Fig 6; Boddy *et al.* 2004).

Optimum pH for radial extension of heterokaryons of all three *Hericium* species was pH 5.5 (7.5–8.7 mm $d^{-1}$), with no growth at pH 2 nor pH 8 (Fig 7; Boddy *et al.* 2004). This correlates with the pH of beech wood, which is around pH 5 (Gray 1958). Again the effect of water potential on extension rate on agar was similar to that for many wood-decay basidiomycetes (Griffin 1981; Boddy 1983; Koske & Tessier 1986), and extension rates decreased with decreasing water potential. *H. erinaceus* was unable to grow at −5 MPa (obtained with KCl), but both *H. cirrhatum* and *H. coralloides* still showed limited growth at this low water potential (Fig 8).

We are unaware of any studies of mycelial extension rates of *Hericium* species in wood, but *H. coralloides* and *H. erinaceus* are easily established in wood by inserting pre-colonised dowels into freshly felled logs and indeed the latter is cultivated commercially for culinary and medicinal use (Suzuki & Mizuno 1997; Stamets 2000).

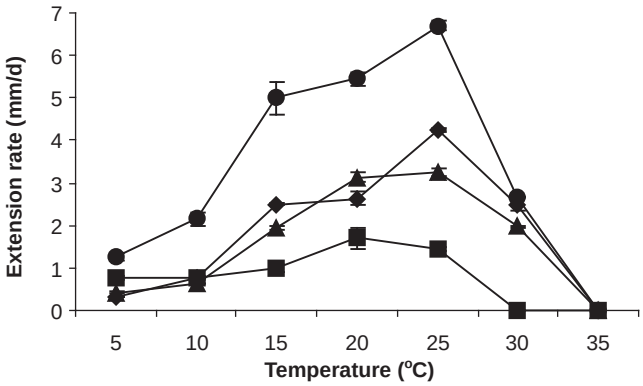


**Fig 6 – Effect of temperature on linear extension rate, on 2 % malt agar, of one isolate of: ▲, *H. cirrhatum*; ◆, *H. coralloides*; ●, *H. erinaceus*; and ■, *H. flagellum*. Bars are 95 % confidence interval, and where no error bar is shown errors are within the size of the symbol. From Boddy *et al.* (2004).**

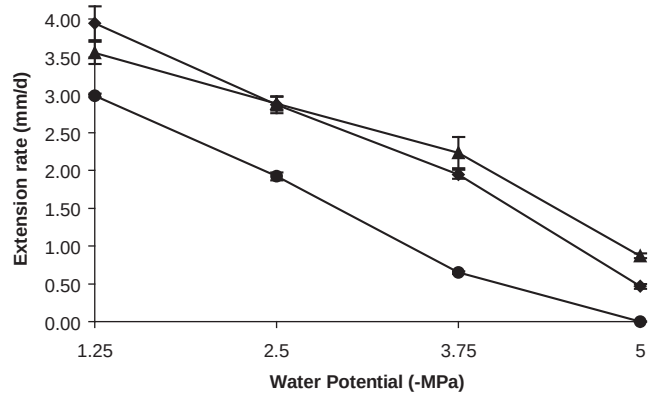


**Fig 8 – Effect of solute potential exerted by KCl, on 2 % malt agar at 20 °C, on radial extension rate of one isolate each of: ▲, *H. cirrhatum*; ◆, *H. coralloides*; and ●, *H. erinaceus*. Error bars are 95 % confidence interval, and where no error bar is shown errors are within the size of the symbol. From Boddy *et al.* (2004).**

There is little information on the size of individual decay columns of these species in dead wood because destructive analysis of wood on which they are fruiting, or have recently fruited, has rarely been performed. However, a study carried out on Turkey oak (*Quercus cerris*) logs lying on the ground in wood pasture revealed large areas colonised by *H. cirrhatum* (Fig 9; Parfitt *et al.* 2005). Interestingly, this species was detected, using PCR specific primers, from regions where it was predicted to be found, based on patterns of staining and decay, but from which it was not isolated by the standard technique of plating wood chips onto malt agar. The implication was that in these regions, hyphae were sparse in the middle of decay columns or the fungus was in the process of being replaced. It had fruited on these logs *in situ* the year before, and fruited again when the sliced and sampled wood was incubated in the lab. The fact that fruiting of the species appears usually only to occur once (very occasionally two or three times) on a piece of wood, (Marren & Dickson 2000; AM Ainsworth pers. obs.) might, then, be because it fruits immediately prior to its mycelial replacement.

How the tooth fungi establish in the field is not known. However, there is evidence from use of PCR primers specific for *Hericium* spp. on DNA extracted from functional sapwood of a range of angiosperm trees (ash, beech, elder (*Sambucus nigra*), hazel (*Corylus avellana*) and field maple (*Acer campestris*)) that these fungi may be latently present as endophytes, only

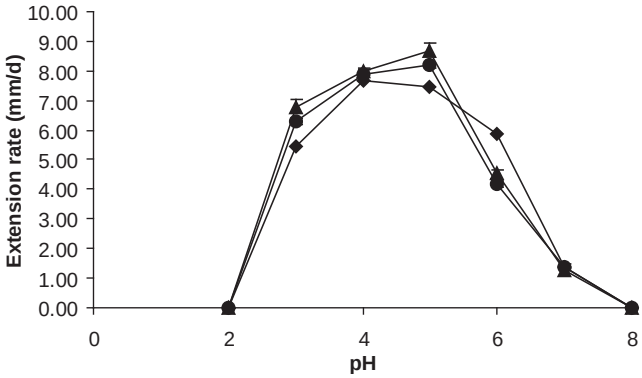


**Fig 7 – Effect of pH on mean linear extension rate, on malt agar at 20 °C, of a single isolate of: ▲, *H. cirrhatum*; ◆, *H. coralloides*; and ●, *H. erinaceus*. Error bars are 95 % confidence interval, and where no error bar is shown errors are within the size of the symbol. From Boddy *et al.* (2004).**

**Fig 9 – Section through a Turkey oak (*Quercus cerris*) log (approximately 58 cm diameter). The presence of *Hericium cirrhatum* was detected by PCR 'specific' primers in the regions outside of the line. The arrow indicates the position of fruit bodies a few months prior to analysis. © Juliet Hynes.**

growing overtly as mycelia when wood begins to dry (Parfitt *et al.* 2010). Attempts to inoculate *H. coralloides* into functional sapwood of standing trees were largely unsuccessful. When trunks were harvested 52 months after autumn inoculation, the fungus was only detected in a few samples, up to 19 cm from the inoculation point of a single dowel, (using PCR primers specific to *H. coralloides*/*H. erinaceus* but that could not distinguish between the two) with no detectable recovery from around the other dowels (Crockatt 2008). Consistent with this is a report of *H. erinaceus* fruiting from a strip of heartwood prematurely exposed on a wounded young plantation oak in the New Forest (AM Ainsworth pers. obs.) suggesting that the fungus can establish in younger trees and produce fruit bodies. However, fruiting is not often seen on young trees due to the relative scarcity of fractures, wounds or otherwise exposed areas of dead wood on such trees.

## Mating

The majority (90 %) of Agaricomycetes have diaphoromictic mating systems that favour out-breeding (i.e. outcrossing between non-siblings) (Esser 1967; Hibbett & Donoghue 2001). The significance of breeding systems is likely to be greater for species that produce few fruit bodies than for those whose fruit bodies are common, as it is more likely that primary mycelia from the former will encounter siblings rather than non-siblings, leading over generations to a loss of heterozygosity in populations. Loss of heterozygosity may lead to decreased fitness and even extinction as seen with the Glanville fritillary butterfly *Melitaea cinxia* (Saccheri *et al.* 1998). On the other hand, non-outcrossing (clonally reproducing) systems may be selected for in rare fungi and other fungi at edges of their range, since each primary mycelium can sporulate in isolation. This is often seen in plants where there is inbreeding pressure in small populations at the edge of their range, often with strong selection against incompatibility mechanisms that would normally ensure out-breeding, frequently resulting in reduced genetic diversity (Arnaud-Haond *et al.* 2006; Beatty *et al.* 2008; Leimu *et al.* 2010). Pairing basidiospore-derived cultures from the same and from different fruit bodies indicates that *H. coralloides* and *H. erinaceus* in North America have bifactorial (tetrapolar) mating systems (Ginns 1985) and there are also indications that *H. coralloides* within Europe is bifactorial (Hallenberg 1983). There is also some evidence to support *H. coralloides* and *H. erinaceus* having bifactorial mating systems in the UK, with 25–31 % compatibility of *H. coralloides* siblings and 22 % compatibility of *H. erinaceus* siblings in pairings on malt agar (Crockatt 2008). The mating system of *H. cirrhatum* remains unresolved: although clamp connections were seen on pairings between non-sibling primary mycelia in almost 100 % of pairings but less in sibling pairings, in the UK it was not possible to distinguish mating types, and the data remain confusing (Crockatt 2008).

## Interspecific interactions in wood

The ability of species to establish in wood that is already colonised by other species, and to remain in wood for extended periods when other species are trying to invade, depends on the combative ability of the fungus. In interactions on agar between *H. cirrhatum*, *H. coralloides* and *H. erinaceus* against 20 other wood-decay fungi from beech, covering a range of ecological strategies, the tooth fungi deadlocked with or replaced over half of the antagonists (Table 3; Wald *et al.* 2004; Crockatt *et al.* 2008). *H. erinaceus* was slightly more combative than the other two species. The *Hericium* species were only rarely replaced by heart rot fungi or primary colonisers latently present in sapwood. Moreover, they were often able to maintain territory against fungi considered to be secondary colonisers of standing dead wood, e.g. *T. versicolor* and *Stereum hirsutum*, though not against *Bjerkandera adusta*. The prolonged occupancy of dead wood by *Hericium* species is likely to be at least partly related to their combative ability. In contrast to outcomes against early colonisers, the combative late stage fungi such as the cord-former *Hypholoma fasciculare* replaced the tooth fungi both in agar and in wood. *H. fasciculare* was, however, notably less successful under reduced water potential and reduced oxygen/elevated $CO_2$. Reduced water potential affected the outcome of many interactions slightly favouring *H. cirrhatum* and *H. erinaceus*.

Mycelial extension rates of *H. cirrhatum* and *H. erinaceus* often increased in the presence of antagonists in agar. Such stimulation might be viewed as indicating that the fungus is doing well, especially in pairings in which it wins, though it could equally indicate an escape response in the presence of a competitor. For example, extension rate was increased against *H. fasciculare*, a combative fungus. There was also evidence that the tooth fungi produced volatile or diffusible compounds that were inhibitory to antagonists on agar and may be even more effective in the large volumes of wood that they presumably occupy in standing trees.

With rare fungi that may have an extended homokaryotic phase, the combative ability of both homokaryons and heterokaryons is important to establishment and survival within wood. In agar culture, both homokaryons and heterokaryons of *H. coralloides* are combative, the former slightly more so than the latter (Crockatt *et al.* 2008). Although combative ability of homokaryons of *H. cirrhatum* and *H. erinaceus* has not been examined, the heterokaryons are combative in agar and wood, thus, combativeness is unlikely to contribute to the apparent rarity of these species.

## Conservation management

Management of all three of these species must largely depend on conservation of trees and sites currently known to harbour fruiting individuals and, in the longer term, ensuring continuity of tree species and conditions which favour fruiting, spore production and subsequent colonisation. Practical measures range from retention of individual branches fallen or cut from known occupied trees, to the enhancement of ecological continuity at the landscape scale. It is essential to leave as much large diameter dead wood *in situ* as possible, but relocation of detached limbs to nearby woodland is probably the most pragmatic option in highly formal settings such as gardens and vistas.

For *H. erinaceus*, re-erecting fallen colonised trunks after wind throw or fracture has been suggested as a management

**Table 3 – Range of outcomes of interactions between tooth fungi and other wood-decay basidiomycetes and ascomycetes (asterisk). Summarized from Wald *et al.* (2004).**

| | Ecological role | *Hericium cirrhatum* | *Hericium coralloides* | *Hericium erinaceus* |
|---|---|---|---|---|
| *Biscogniauxia nummularia** | P | pr r | pr, r | r |
| *Diatrype disciformis** | P | pr | pr | D, pr |
| *Eutypa spinosa** | P | PR, D | PR, D, pr | D, r |
| *Antrodiella semisupina* | L | pr | pr | |
| *Bjerkandera adusta* | S | R, PR | R, PR | R, PR |
| *Ceriporiopsis gilvescens* | L | R | R | |
| *Chondrostereum purpureum* | P | PR | PR | |
| *Coniophora puteana* | P | R | R | |
| *Fomes fomentarius* | H | r, pr, PR | r, pr, PR, D | D, r, pr |
| *Ganoderma applanatum* | H | r | r, pr | r |
| *Hericium cirrhatum* | | | D | D, pr |
| *Hericium coralloides* | | D | | D |
| *Hericium erinaceus* | | D, pr | D | |
| *Hypholoma fasciculare* | C | R, PR, D | R, PR, D | PR, D |
| *Inonotus obliquus* | H | r | r, pr | D, r, pr |
| *Lycoperdon pyriforme* | L | D, PR | D, R, pr | D, pr |
| *Peniophora lycii* | A | PR, D | | PR, D |
| *Phanerochaete velutina* | C | PR | PR | |
| *Phlebia livida* | L | PR, D | PR | |
| *Phlebia radiata* | S | PR, R, D, pr | R, PR, pr | R, PR, D, pr, r |
| *Piptoporus quercinus* | H | D, pr | D(1), pr(3) | pr |
| *Stereum hirsutum* | S | D | D, pr | D |
| *Stereum gausapatum* | P | D, PR | D, PR | |
| *Stereum rugosum* | P | D, pr | pr | |
| *Trametes versicolor* | S | R, PR, D | D, PR | R, D, pr |
| *Vuilleminia comedens* | P | D, pr | r | |

Abbreviations for ecological roles: A, attached branch, dry; C, late secondary-colonising cord-former; H, heart-rotter; L, other late secondary-colonisers; P, primary colonisers latently present in stems and attached branches; S, early secondary coloniser.
Abbreviations for outcome of interactions: D, deadlock; R, replacement of *H. cirrhatum*, *H. coralloides* or *H. erinaceus* by opponent; PR, partial replacement of *H. cirrhatum*, *H. coralloides* or *H. erinaceus* by opponent; r, replacement of opponent; pr, partial replacement of opponent. Variation in outcome for any combination resulted from different abiotic regimes and differences between strains and replicates.

option. The aim is to try to maintain the historically favourable microclimatic regime for as long as possible and to delay replacement by combative fungi establishing from mycelial cord systems within the woodland litter. To this end, a re-erection experiment was set up in 2001 involving strapping a fallen section of occupied trunk to a neighbouring healthy beech tree (Borges & Rotheroe 2002). However, this resulted in no further fruiting of *H. erinaceus*, and drying of the largely decorticated trunk, whereas another part of the same fallen trunk which was left in place, did not dry out and went on to produce fruit bodies of rarely-recorded species, though none of *Hericium*. Field observations suggest that when *H. erinaceus*-colonised wood breaks and a main branch falls, *H. erinaceus* can simultaneously fruit on the fracture faces created on the standing tree and fallen branch, implying that fruiting is not directly related to location but to exit routes, aeration and/or drying. Moreover, *H. erinaceus* can continue to fruit on fallen wood that has been detached from the canopy for a very long time and lost almost all structure. Therefore current evidence does not support the re-erection of fallen limbs as a beneficial conservation measure for *H. erinaceus*.

If *Hericium* spp. are confirmed as latent invaders, their mycelia could be present for many years within the central regions of trees and remain undetected until there is a fruiting opportunity. This could present a management issue at *Hericium* sites with multi-purpose objectives, such as beech forestry on sites with statutory conservation designations, because of the difficulty in selecting exactly which trees should be by-passed by the forester at harvest time and left "for the fungi".

### *Conclusions and research priorities*

The three *Hericium* spp. have broadly similar ecological preferences, but there is some degree of niche differentiation (Table 1). This may translate into differences, albeit subtle, in their conservation strategies. In nature, though they fruit infrequently, and presumably late in their life cycle, *Hericium* species produce large quantities of spores. However, germination success on artificial media was usually low, and poor germination may be a factor contributing to their rarity. There is evidence that they are sometimes latently present in functional sapwood of some angiosperm trees (Parfitt *et al.* 2010) but, as for other latent wood-decay fungi (Boddy 2001), where and when they enter wood and their precise mode of entry is unclear. Furthermore, the process and success rate of switching over from existing as latent propagules to overt establishment as mycelium is also unknown. Although *Hericium* spp. have not been recorded in previous investigations of decay column establishment by early colonisers in beech (Chapela & Boddy 1988; Hendry *et al.* 2002), this work relied on morphological identification methods alone. Interestingly,

unidentified basidiomycetes were obtained in these studies, but they were not sequenced at the time and it is possible that *Hericium* spp. were present.

The mechanism(s) whereby *Hericium* spp. basidiospores colonise and become established in standing trees is clearly a research priority (Table 2). They are easily established in uncolonised felled wood by inoculating logs with pre-colonised dowels. However, we had no success inoculating live standing trees with *H. coralloides*, but this does not necessarily imply that they do not naturally establish in standing trees, as discussed above. It would be valuable to determine how frequently and under what conditions the latent propagules detected by PCR within sapwood are able to establish as overt decay columns, and to understand how these fungi initiate colonisation *via* basidiospores (Table 2).

Ecological research is urgently needed to study the population structure of all three species (Table 2). From a conservation status assessment standpoint, it would be particularly useful to know the average number of genetically and spatially distinct *Hericium* individuals (genets and ramets) that occur within an occupied tree. Further data on distribution of individuals of these and other wood-inhabiting fungi would potentially increase the criteria available for carrying out Red Data Listing and lead to some species status reassessments. It is worth bearing in mind in this respect that even populations of up to 10 000 mature individuals (defined as ramets by IUCN) can be classed as "vulnerable" on Red Lists provided they satisfy standard IUCN criteria (IUCN 2001, 2003).

While rarity has been mentioned frequently above, whether *Hericium* species are rare remains a moot point. Their fruit bodies are hard to miss, so fruit bodies are probably not under recorded. However, since they only fruit infrequently, there are likely to be more undisclosed individuals in nature. *H. erinaceus* is unlikely to become extinct because it is cultivated for production of edible fruit bodies. However, the availability of *Hericium*-growing kits raises the possibility that introduced (alien) genotypes might soon (already) be competing with native stock in the field. A practical priority for a legally protected species such as *H. erinaceus* is a comparative DNA fingerprinting study of native stock and mycelia sold in kits. This could enable detection of exotic *H. erinaceus* and facilitate tracking of its distribution trends, providing timely information on any potential displacement of the native population.

We now know a lot about the ecology of the three *Hericium* species, and the remaining lacunae have been highlighted (Table 2). Filling these knowledge gaps will provide models of *Hericium* populations, which might make it easier to identify the crucial stages for maintaining viable populations of these species, and also of other rare and threatened wood-decay fungi. Investigating the numbers of genets per tree is the highest priority, as this will improve conservation assessments for the species. Population approaches are essential for the integration of fungal status assessments and IUCN criteria.

## Acknowledgements

Funding: Cardiff University, Forestry Commission England, Natural England. Access and permission to work on their sites: City of London, Crown Estate (Windsor), Forestry Commission England. Assistance with records and recording: Hants Fungus Recording Group, P. Kirk (BMS Database Manager), E.E. Green, A. Lucas, and all those recorders who kindly shared their data.

Text improvement: Brian Spooner and two anonymous referees. Advice on loss of heterozygosity and inbreeding in animals and plants: Mike Bruford and Stuart Davies.

## REFERENCES

Adams TJH, Williams END, Todd NK, Rayner ADM, 1984. A species specific method of analysing populations of basidiospores. *Transactions of the British Mycological Society* **82**: 359–361.

Ainsworth AM, 2004. *BAP Fungi Handbook* (English Nature Research Report number 600). English Nature, Peterborough.

Ainsworth AM, 2008. Surveys, strongholds and revised red listing of the legally protected wood-inhabiting fungi oak polypore *Piptoporus quercinus* and bearded tooth *Hericium erinaceus*. Unpublished report. Natural England, Peterborough.

Arnaud-Haond S, Teixeira S, Massa SI, Billot C, Saenger P, Coupland G, Duarte CM, Serrao A, 2006. Genetic structure at range edge: low diversity and high inbreeding in Southeast Asian mangrove (*Avicennia marina*) populations. *Molecular Ecology* **15**: 3315–3525.

Beatty GE, McEnvoy PM, Sweeney O, Provan O, 2008. Range-edge effects promote clonal growth in peripheral populations of the one-sided wintergreen *Orthilia secunda*. *Diversity and Distributions* **14**: 546–555.

Boddy L, 1983. The effect of temperature and water potential on the growth rate of wood-rotting basidiomycetes. *Transactions of the British Mycological Society* **80**: 141–149.

Boddy L, 2001. Fungal community ecology and wood decomposition processes: from standing tree to complete decay of coarse woody debris. In: Jonsson BG, Kruys N (eds), *Ecology of Woody Debris in Boreal Forests*, vol. 49. Ecological Bulletins, pp. 43–56.

Boddy L, Wald P, 2003. Creolophus (=Hericium) cirrhatus, Hericium erinaceus *and* H. coralloides *in England* (English Nature Research Report number 492). English Nature, Peterborough.

Boddy L, Wald PM, Parfitt D, Rogers HJ, 2004. *Preliminary Ecological Investigation of Four Wood-Inhabiting Fungi of Conservation Concern – oak polypore* Piptoporus quercinus (=Buglossoporus pulvinus) *and the tooth fungi* Hericium/Creolophus *spp* (English Nature Research Report number 616). English Nature, Peterborough.

Borges C, Rotheroe M, 2002. *Managing Land with Fungi in Mind*, vol. 10. Enact Magazine (English Nature). 17–22.

Chapela IH, Boddy L, 1988. Fungal colonisation of attached beech branches. 2. Spatial and temporal organisation of communities. *New Phytologist* **110**: 47–57.

Coates D, Rayner ADM, 1985. Fungal population and community development in cut beech logs. 1. Establishment via the aerial cut surface. *New Phytologist* **101**: 153–171.

Crockatt ME, 2008. Ecology of the rare Oak polypore *Piptoporus quercinus* and the tooth fungi *Hericium cirrhatum*, *H. coralloides* and *H. erinaceus* in the UK. PhD thesis, Cardiff University, UK.

Crockatt ME, Ainsworth AM, Parfitt D, Rogers HJ, Boddy L, 2007. Why are the tooth fungi *Hericium cirrhatum*, *H. coralloides* and *H. erinaceus* rare? In: *World Conference on the Conservation and Sustainable Use of Wild Fungi*. Junta de Andalucia, Cordoba, Spain, pp. 116–118.

Crockatt ME, Pierce GI, Camden RA, Newell PN, Boddy L, 2008. Homokaryons are more combative than heterokaryons of *Hericium coralloides*. *Fungal Ecology* **1**: 40–48.

Dahlberg A, Croneborg H, 2003. *33 Threatened Fungi in Europe* (T-PVS [2001] 34 rev. 2). Complementary and revised information on candidates for listing in Appendix I of the Bern Convention. Swedish EPA & ECCF, Uppsala.

Dahlberg A, Genney DR, Heilmann-Clausen J, 2010. Developing a comprehensive strategy for fungal conservation in Europe: current status and future needs. *Fungal Ecology* **3**: 50–64.

Edman M, Gustafsson M, 2003. Wood-disk traps provide a robust method for studying spore dispersal of wood-decaying basidiomycetes. *Mycologia* **95**: 553–556.

Esser K, 1967. Die Verbreitung der Incompatibilitat bei Thallophyten. *Handbuch der Pflanzenphysiologie* **18**: 321–343.

Evans S, Henrici A, Ing B, 2006. The red data list of threatened British fungi (preliminary assessment). http://www.fieldmycology.net/ (Accessed 10.03.08.).

Garbelotto MM, Lee HK, Slaughter G, Popenuck T, Cobb FW, Bruns TD, 1997. Heterokaryosis is not required for virulence of *Heterobasidion annosum*. *Mycologia* **89**: 92–102.

Gärdenfors U (ed), 2010. *Rödlistade arter i Sverige 2010 – the 2010 red list of Swedish species*. ArtDatabanken, Uppsala.

Ginns J, 1984. *Hericium coralloides* N. amer auct (*Hericium americanum* sp nov) and the European *Hericium alpestre* and *Hericium coralloides*. *Mycotaxon* **20**: 39–43.

Ginns J, 1985. *Hericium* in North America: cultural characteristics and mating behaviour. *Canadian Journal of Botany* **63**: 1551–1563.

Gray VR, 1958. The acidity of wood. *Journal of the Institute of Wood Science* **1**: 58–64.

Guglielmo F, Bergemann SE, Gonthier P, Nicolotti G, Garbelotto MM, 2007. A multiplex PCR-based method for the detection and early identification of wood rotting fungi in standing trees. *Journal of Applied Microbiology* **103**: 1490–1507.

Griffin DM, 1981. Water and microbial stress. *Advances in Microbial Ecology* **5**: 91–136.

Gyosheva MM, Fakirova VI, Denchev CM, 2000. Red list and threat status of Bulgarian macromycetes. *Historia Naturalis Bulgarica* **11**: 139–145.

Hallenberg N, 1983. *Hericium coralloides* and *H. alpestre* (basidiomycetes) in Europe. *Mycotaxon* **18**: 181–189.

Hallenberg N, Küffer N, 2001. Long-distance spore dispersal in wood-inhabiting Basidiomycetes. *Nordic Journal of Botany* **21**: 431–436.

Hansen L, Knudsen H, 1997. *Nordic Macromycetes*, vol. 3. Nordsvamp, Copenhagen, Denmark.

Hendry SJ, Boddy L, Lonsdale D, 2002. Abiotic variables effect differential expression of latent infections in beech (*Fagus sylvatica*). *New Phytologist* **155**: 449–460.

Hibbett DS, Donoghue MJ, 2001. Analysis of character correlations among wood decay mechanisms, mating systems, and substrate ranges in homobasidiomycetes. *Systematic Biology* **50**: 215–242.

IUCN, 2001. *IUCN Red List Categories and Criteria*. Vers 3.1. IUCN Species Survival Commission. IUCN, Gland, Switzerland. Cambridge, UK.

IUCN, 2003. *Guidelines for Application of IUCN Red List Criteria at Regional Levels*. Vers 3.0. IUCN Species Survival Commission. IUCN, Gland, Switzerland. Cambridge, UK.

Kauserud H, Saetre GP, Schmidt O, Peacock C, Schumacher T, 2006. Genetics of self/non-self in *Serpula lacrymans*. *Fungal Genetics and Biology* **43**: 503–510.

Koske RE, Tessier B, 1986. Growth of some wood and litter-decay basidiomycetes at reduced water potential. *Transactions of the British Mycological Society* **86**: 156–158.

Leimu R, Vergeer P, Angeloni F, Ouborg NJ, 2010. Habitat fragmentation, climate change, and inbreeding in plants, 2010. *Annals of the New York Academy of Sciences* **1195**: 84–98.

Legon NW, Henrici A, 2005. *Checklist of the British and Irish Basidiomycota*. Royal Botanic Gardens, Kew, UK.

Marren P, Dickson G, 2000. British tooth-fungi and their conservation. *British Wildlife* **11**: 401–409.

Merrill W, 1970. Spore germination and host penetration by heartrotting hymenomycetes. *Annual Review of Phytopathology* **8**: 281–300.

Parfitt D, Dockrell D, Hunt J, Rogers HJ, Boddy L, 2010. Do all trees carry the seeds of their own destruction? PCR reveals numerous wood decay fungi latently present in sapwood of a wide range of angiosperm trees. *Fungal Ecology* **3**: 338–346.

Parfitt D, Hynes J, Rogers HJ, Boddy L, 2005. New PCR assay detects rare tooth fungi in wood where traditional approaches fail. *Mycological Research* **109**: 1187–1194.

Rayner ADM, Boddy L, 1988. *Fungal Decomposition of Wood: its biology and ecology*. John Wiley and Sons, Chichester.

Redfern DB, Pratt JE, Gregory SC, Macaskill GA, 2001. Natural infection of Sitka Spruce thinning stumps in Britain by spores of *Heterobasidion annosum* and long-term survival of the fungus. *Forestry* **74**: 53–71.

Saccheri I, Kuussaari M, Kankare M, Vikman P, Fortelius W, Hanski I, 1998. Inbreeding and extinction in a butterfly metapopulation. *Nature* **392**: 491–494.

Stalpers JA, 1996. *The Aphyllophoraceous Fungi II. Keys to the Species of the* Hericiales. *Studies in Mycology 40*. Centraalbureau voor Schimmelcultures, Baarn and Delft.

Stamets P, 2000. *Growing Gourmet and Medicinal Mushrooms*. Ten Speed Press, Berkeley, California.

Stenlid J, 1994. Homokaryotic *Heterobasidion annosum* mycelia in Norway Spruce stumps. In: Johansson M, Stenlid J (eds), *Proceedings of the 8th International Conference on root and butt rots*. Swedish University of Agricultural Sciences, Uppsala, Sweden, pp. 249–253.

Suzuki C, Mizuno T, 1997. Cultivation of Yamabushitake (*Hericium erinaceum*). *Food Reviews International* **13**: 419–421.

Wald PD, Pitkänen S, Boddy L, 2004. Interspecific interactions between the rare tooth fungi *Creolophus cirrhatus*, *Hericium erinaceus* and *H. coralloides* and other wood-decay species in agar and wood. *Mycological Research* **108**: 1447–1457.

Wicks D, 1999. Survey of the New Forest for tooth fungi *Hericium erinaceum* and *Hericium coralloides*. Unpublished report. Hampshire and Isle of Wight Wildlife Trust Ltd.